Madhav Jadav

Experienced Blockchain Developer with a passion for Rust and Solana
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Professional Summary

STEM graduate (B.Tech-M.Tech Integrated Computer Science and Engineering, National Forensic Sciences University Delhi, 2021–2026) with 18 months project-based software engineering experience in backend development, REST APIs, and distributed systems for financial markets. Strong computer science fundamentals in data structures, algorithms, system design, and object-oriented and functional programming in Rust, Go, Python, Java, C++, C#, and JavaScript (Node.js). Built low-latency REST APIs serving 5K–10K requests per second with under 12ms latency, containerized microservices with automated testing and CI/CD, and a custom blockchain in Rust (SHA-256, Proof-of-Work) for digital asset custody logging. Experienced across full software development lifecycle: design, development, integration, automated testing, and maintenance in fast-moving release cycles. Collaborated with cross-functional stakeholders including quants, product, algo developers, infrastructure, and support. Genuine interest in crypto, digital assets, trading, execution and order management systems (OMS/EMS), portfolio and risk management. Madhav Jadav, a 2026 B.Tech-M.Tech graduate with a strong background in backend engineering and blockchain technology, is a strong candidate for the Smart Contract Engineer role at Saifuro, where he can leverage his skills in Rust, Go, Python, and SQL to contribute to the company'thre core escrow and verification systems.

Technical Skills

Languages: Rust, Go (Golang), Python, Java, C++, C#, JavaScript/TypeScript (Node.js), SQL
Backend and APIs: REST APIs, Microservices, Event-Driven Architecture, System Design, gRPC, GraphQL, FastAPI, Actix-web, Node.js, JWT, RBAC, Single API Integration
Data and Infrastructure: PostgreSQL, MongoDB, Redis, Apache Kafka, Docker, Kubernetes, Helm, GitHub Actions (CI/CD), AWS EC2, MLflow
Engineering Practices: Software Development Lifecycle (SDLC), Design, Development, Integration, Automated Testing, Maintenance, Agile, Fast-Moving Release Cycle, Continuous Improvement, Stakeholder Collaboration, Code Review
FinTech and Digital Assets: Blockchain, SHA-256, Proof-of-Work, Digital Assets, Crypto, Trading, Execution, Order Management System (OMS/EMS), Portfolio Management, Risk Management, Market Connectivity, Secure SDLC, OWASP Top 10
Computer Science Fundamentals: Data Structures, Algorithms, Object-Oriented Programming, Functional Programming, Distributed Systems, Computer Networks, Operating Systems

Education

National Forensic Sciences University (NFSU), Delhi, India Nov 2021 – May 2026
B.Tech-M.Tech Integrated Computer Science and Engineering (Cyber Security) – STEM *CGPA: 7.5/10.0*

- Graduated 2026. Coursework: Data Structures, Algorithms, System Design, DBMS, Operating Systems, Computer Networks, Distributed Systems, Cryptography, OOP.
- Programming in Rust, Go, Python, Java, C++, JavaScript (Node.js). Strong analytical and problem-solving skills.

Premdhara Shishu Vihar High School, Gandhinagar, India – HSC (Class XII) 2021

- State Government, Gujarat. Physics, Chemistry, Maths.

Modern High School, Gandhinagar, India – SSC (Class X) 2019

Software Engineering Experience

Backend Software Engineer (Project-Based) – REST APIs, Microservices, Digital Assets Jan 2023 – Present
Self-Directed Engineering, Open Source and Hackathons | Delhi, India

- Develop and enhance backend software across full technology stack: design, development, integration, automated testing, and maintenance in fast-moving release cycles.
- Build and maintain features with business stakeholders: low-latency REST APIs serving 5K–10K requests per second with under 12ms latency using FastAPI, Redis caching, and PostgreSQL.
- Deliver trading-adjacent solutions: custom blockchain ledger in Rust (Actix-web, SHA-256, Proof-of-Work) modelling immutable execution logging, custody, and risk management workflows for digital assets.
- Work with cross-functional collaborators (product, algo/ML developers, infrastructure, support) to ship containerized microservices on Docker and Kubernetes with Helm, Kafka event streaming, and GitHub Actions CI/CD (deployment time cut from 2 hrs to 7 mins).

- Leverage new technologies with continuous improvement: gRPC, GraphQL, event-driven design, RBAC with Open Policy Agent, MFA/JWT authentication with 99.99% uptime and zero unauthorized access in penetration tests.

Hackathon Engineer and CTF Competitor – Algo Development, Rapid Prototyping 2023 – 2025
Unstop, Devfolio, RedFox Security, SRM University | India

- Competed in 10+ hackathons and CTFs; built secure backend and AI/ML prototypes under deadlines, collaborating with product-minded stakeholders.

Projects

RhinoChain: Secure Biometric Blockchain System | Rust, Actix-web, SHA-256, Proof-of-Work, REST API Oct 2024 – Jan 2025
github.com/jadavmadhavkumar/rhino_chain

- Developed custom blockchain in Rust for immutable storage of hashed digital assets using SHA-256 and Proof-of-Work consensus – directly relevant to digital asset trading, execution logging, and risk management infrastructure.
- Built and documented single REST API for block addition, chain validation, and real-time updates; modular architecture with persistent storage and automated testing for cryptographic integrity.
- Applied system design, data structures, and algorithms for tamper-proof, privacy-preserving backend workflows and market connectivity concepts.

Secure Authentication Microservice | Node.js, JWT, MFA, RBAC, Kubernetes, Helm Sep 2023 – Dec 2023

- Engineered stateless backend microservice handling 10K RPS with JWT + TOTP MFA; 0% unauthorized access in penetration tests; 99.99% uptime with auto-scaling.
- Implemented RBAC with dynamic policies (Open Policy Agent) for 5+ roles; deployed via Docker, Kubernetes, Helm with full SDLC: design, integration, automated testing, maintenance.

ML-Powered Backend Engine (FastAPI + Redis) | Python, Scikit-learn, FastAPI, Redis, PostgreSQL Mar 2024 – Jun 2024

- Developed real-time backend classifier (99.2% accuracy on 1M records) exposed via FastAPI with Redis caching – under 12ms latency at 5K QPS; integrated into SSO portal, reducing weak inputs by 73%.
- Demonstrates low-latency API design, performance optimization, and continuous improvement applicable to execution and portfolio systems.

Real-Time Streaming Backend | Python, OpenCV, Kafka, Docker, GitHub Actions Dec 2022 – Mar 2023

- Architected event-driven streaming pipeline (30 FPS, 98.7% accuracy) with Apache Kafka and Docker; automated CI/CD reducing deployment from 2 hrs to 7 mins.

DcGAN Image Synthesis API | PyTorch, MLflow, AWS EC2 Oct 2023 – Jan 2024

- Trained DcGAN (generator loss 0.014, FID improved 42%) with MLflow tracking; deployed scalable REST API on AWS Spot Instances, cutting inference cost 68%.

Achievements

- **First Prize – Verge Technical Fest, SRM University (2025):** Top position for innovation, technical excellence, and problem-solving under deadlines. LinkedIn
- **Second Prize – CyberStorm CTF, RedFox Security (2025):** National cybersecurity competition: cryptography, reverse engineering, web exploitation. LinkedIn
- **Top Performer – 10+ National Hackathons, Unstop/Devfolio (2023–2025):** Built secure backend and AI/ML prototypes for rapid release cycles.

Certifications

- **Introduction to Generative AI** – Google Cloud Skills Boost (2023)
- **Cyber Apocalypse CTF** – Hack The Box, Global CTF (2023, 2024): web exploitation, cryptography, reverse engineering; secure SDLC practice.
- **Build Greeting App with React JS** – Coding Ninjas (2023)
- **Tata Data Visualization (TCS Forage)** – Executive dashboards and client-facing insights (Jan 2025)